Correct-Response Novelty in Full-Weight LLM Evolution

Libor "Burny" Burian

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Abstract

We test an adaptation of historical response novelty to full-weight Gaussian evolution strategies on witnessed arithmetic problems. The comparison uses one paired training seed and an untrained base model. The directional improvement hypothesis was not supported in this seed: main accuracy did not exceed reward-only ES. Problem-resampling intervals exclude training-seed variability. All tables are generated from persisted evaluations; incomplete outcomes are labeled.

Context, motivation and introduction

The experiment asks whether exploration among correct answer forms improves held-out task success. ExTra [1] supplies the historical semantic novelty concept; ES at Scale [2] supplies the full-weight optimization reference. Their combination uses population-level ES rather than GRPO, with a different sibling population and task, so it is an adaptation rather than a reproduction. A reward-only ES control is necessary because ES can itself change solution coverage [3].

Background and related work

Novelty-seeking ES was previously studied in control [4]. LLM-specific alternatives include EVOL-RL's within-group novelty [5], DARLING's quality/diversity objective [6], and the model/task coevolution of AC/DC [7]. These use different training or population mechanisms. The repository literature audit distinguishes prompt evolution, artifact search and actual model-parameter training, with code/license verification.

Method

The model is Qwen/Qwen2.5-0.5B-Instruct, revision 7ae557604adf67be50417f59c2c2f167def9a775. Each generation evaluates 30 independent Gaussian candidates (sigma 0.001) on 32 rotating training problems and 16 fixed training probes. Population-z-scored task reward is combined with 1.0 times correctness-gated cosine novelty for the main condition. The archive retains 6 correct embeddings per probe after seeded insertion. These coefficients weight noise directly, without another normalization; effective alpha is 0.0005. FP32 masters reconstruct BF16 candidates afresh and tied tensors are updated once. Prefix regeneration and backpropagation are absent.

Qwen/Qwen2.5-0.5B-Instruct: 24 layers, width 896, 14 query heads and 2 key/value heads; 494,032,768 unique parameter elements in 290 tensors. Architecture and counts come from the pinned pretrained files and the generation-zero checkpoint audit.

For candidate i, Q_i is the mean response reward: exact success receives 1, a legal expression using every supplied number but missing the target receives 0.1, and an invalid expression receives 0. For probe p, n_i,p = I(correct_i,p) [1 - max cosine(e_i,p, z)]. The reference set consists of correct historical responses and correct responses from other candidates on that same probe; self-comparisons are excluded. An empty set has maximum zero. N_i is the probe mean. History is frozen while scoring the whole generation, then correct embeddings are appended in seeded shuffled order and the oldest entries are removed.

Z_i = (Q_i - mean(Q)) / (std_population(Q) + 1e-8), with zero Z for constant Q. W_i = Z_i + 1.0 N_i for the main arm and W_i = Z_i for control. The update is theta_next = theta + (alpha / population) sum_i W_i epsilon_i. Each epsilon_i is independent standard Gaussian noise over every unique parameter tensor. A common positive novelty bonus is retained. Alpha includes the inverse-noise factor; it is not an additional alpha/sigma update. MiniLM is frozen, and both conditions compute the same categories of probe work.

Protocol amendment A replaced combined-reward ranks with source-aligned z-scored quality plus post-normalization novelty. It was published after calibration and partial original-model evaluation were visible, but before either trained condition began. Both arms start from the original pretrained weights using the amended rule from their first generation. The original registration, changed configuration, code revisions and unchanged absolute deadline are retained in the run record. This ordering follows ES at Scale Algorithm 2 and ExTra Equation 2; it is not evidence that the hybrid outperforms the abandoned rule.

Data, evaluation and preregistered analysis

Synthetic problems have exact rational witnesses. Global number-multiset isolation prevents permutations or alternative targets crossing splits. Only constants supplied in the prompt and allowlisted arithmetic are accepted. Greedy test accuracy at the final scheduled checkpoint is primary. The directional hypothesis is falsified for this pilot if main minus control accuracy is nonpositive. A paired problem bootstrap quantifies conditional uncertainty. Sampled pass@k follows the unbiased subset estimator [8]. Validation cannot select checkpoints. Exact dataset hashes, schedules, timing rules and sampling settings are recorded in reports/preregistration.md and configs/pilot.json.

Dataset counts: train 8192, validation 512, test 2048, hard 1024. Greedy responses allow at most 128 new tokens. The sampled evaluation uses 16 responses on each of 256 fixed test problems, with temperature 1.0, top-p 1.0 and top-k disabled. pass@k = 1 - C(n-c,k)/C(n,k), where c is the number of correct samples. Intervals use 10000 paired resamples, seed 20261107.

Results and claims

The directional improvement hypothesis was not supported in this seed: main accuracy did not exceed reward-only ES.

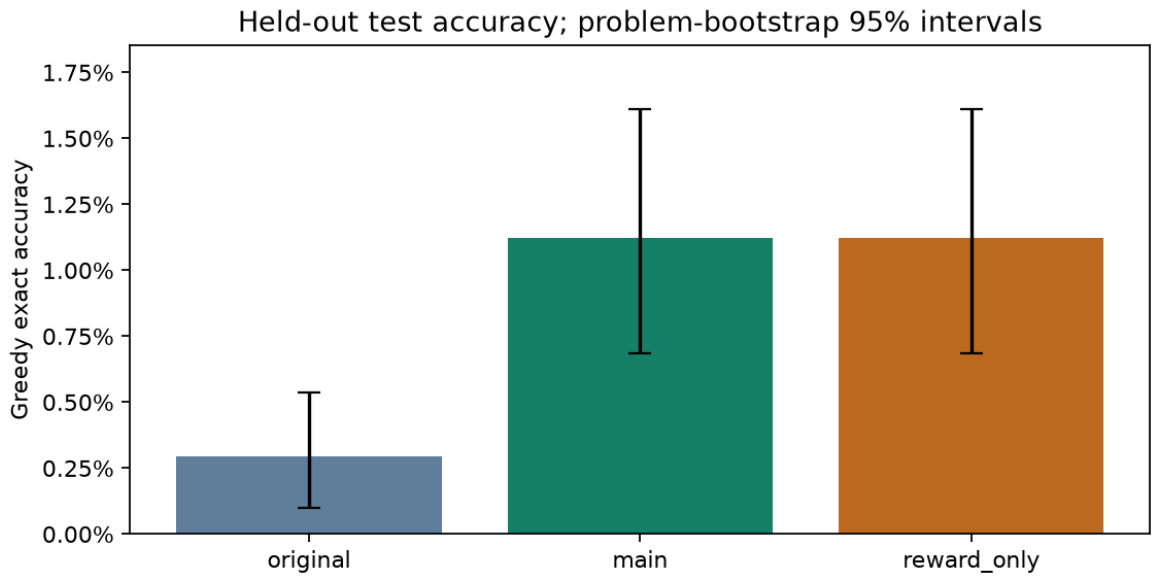
Condition	Test accuracy	95% interval	Hard accuracy	pass@1	pass@8	Status
original	0.29%	0.10% - 0.54%	0.10%	0.12%	0.89%	complete
main	1.12%	0.68% - 1.61%	0.00%	0.27%	2.06%	complete
reward_only	1.12%	0.68% - 1.61%	0.10%	0.22%	1.67%	complete

original: greedy problems test 2048/2048, hard 1024/1024, probes 16/16; sampled complete problems 256/256. main: greedy problems test 2048/2048, hard 1024/1024, probes 16/16; sampled complete problems 256/256. reward_only: greedy problems test 2048/2048, hard 1024/1024, probes 16/16; sampled complete problems 256/256.

Main minus control: 0.000 percentage points; paired 95% interval [-0.244, 0.244], on 2048 paired problems.

Comparison	Difference (pp)	Paired 95% interval (pp)	Problems
Main - reward-only (primary)	0.000	[-0.244, 0.244]	2048
main - original (secondary)	0.830	[0.438, 1.270]	2048
reward_only - original (secondary)	0.830	[0.391, 1.270]	2048

Comparisons with the original checkpoint are descriptive secondary estimates with unadjusted intervals.



Percentile bootstrap intervals can collapse at observed boundaries or when paired outcomes are identical. A zero-width interval does not prove zero population error or equivalence on unseen problems.

Probe performance, validity and resources

Condition	Probe accuracy	Test validity	Hard validity	Train tokens	Eval tokens	Train hours
original	0.00%	29.30%	10.64%	0	385121	0.000
main	0.00%	76.12%	2.05%	798675	217202	1.688
reward_only	0.00%	79.79%	3.52%	814802	229510	1.662

Within-problem expression diversity

Condition	Greedy correct structures	Complete sampled problems	Mean correct structures / sampled problem
original	6	256	0.0156
main	23	256	0.0391
reward_only	23	256	0.0312

Greedy structure counts equal the number of successful test problems because only one answer is generated per problem. The sampled statistic counts distinct correct canonical ASTs within each problem before averaging, including zeros. It measures syntactic diversity, not algebraic or semantic equivalence.

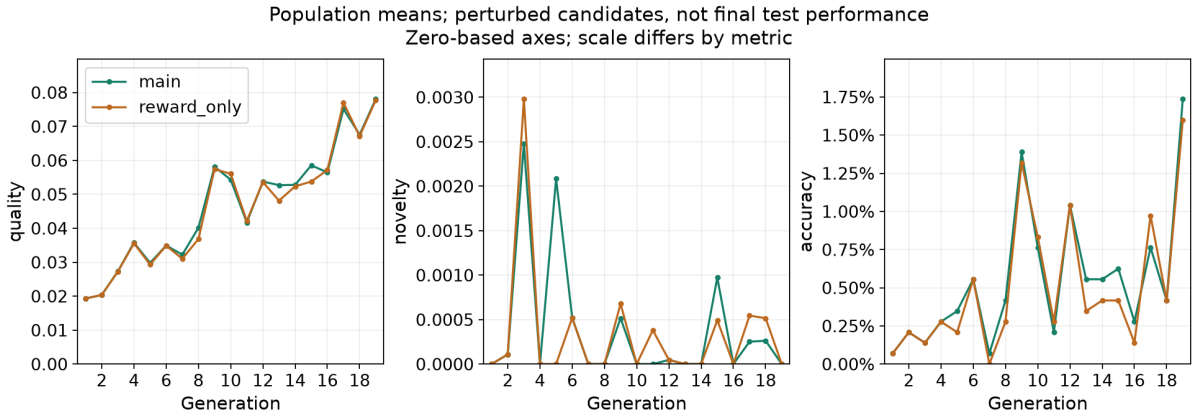
Train tokens count candidates in committed generations; evaluation tokens count persisted evaluation responses. Calibration, validation, interrupted batches and replay costs are excluded from these two table columns. Training hours are wall-clock phase duration and include checkpointing. Probes are training examples, not held-out data.

Did the novelty bonus affect optimization?

Descriptive diagnostic: compare the main arm's Z(Q) + 1.0 N with Z(Q) on its recorded populations. The control computes novelty but applies coefficient zero, so its novelty-induced weight changes are zero. This local comparison does not simulate the counterfactual training trajectory. main: positive novelty in 15 of 570 candidates; changed update weights in 9 of 19 generations (14 candidate weights). reward_only: positive novelty in 13 of 570 candidates; changed update weights in 0 of 19 generations (0 candidate weights).

Training dynamics and output inspection

Population metrics describe perturbed candidates, not the master checkpoint. Reward is the optimization signal; no cross-entropy loss is optimized in this pilot. The interactive result page shows the same first test problems across methods, including failures, without choosing examples by outcome.



Fixed example: original

Use the numbers [49, 11, 13] exactly once each to make 73. You may use +, -, *, / and parentheses. Do not use other numbers or operations. Return only <answer>expression</answer>.
Output: <answer>49 * (11 - 13)</answer>

Fixed example: main

Use the numbers [49, 11, 13] exactly once each to make 73. You may use +, -, *, / and parentheses. Do not use other numbers or operations. Return only <answer>expression</answer>.
Output: <answer> (13 - 49) * 11 </answer>

Fixed example: reward_only

Use the numbers [49, 11, 13] exactly once each to make 73. You may use +, -, *, / and parentheses. Do not use other numbers or operations. Return only <answer>expression</answer>.
Output: <answer> (49 - 11) * (11 + 13) </answer>

Implications and discussion

Both trained checkpoints score higher than the original model on this test set; the main-versus-control comparison determines whether novelty adds a benefit. Both also produce more legal expressions on the three/four-number test problems. Legality alone does not mean the target was reached. Both have lower expression validity on the five-number test problems than the original model, so the validity improvement does not transfer uniformly. This pilot is a controlled objective intervention under one initialization/seed schedule. It can identify a conditional difference or a feasibility failure, but cannot establish a general advantage of novelty training. Any improvement must be checked across additional seeds and separated from probe memorization. A rise in novelty alone, or a selected qualitative example, is insufficient evidence.

Limitations

Training-seed sensitivity is unmeasured. Embedding similarity is a proxy and may reward irrelevant wording; canonical expression trees are an independent syntactic diagnostic, not full algebraic equivalence. Synthetic arithmetic does not measure general language quality. Similar pretraining patterns cannot be excluded. The fixed-probe objective can overfit. Small time budgets, laptop thermal variability and batch-level floating-point behavior limit extrapolation. CPU tests, report generation, audits and backups ran alongside training; elapsed times are descriptive, not isolated systems benchmarks. Secondary endpoints are descriptive; no multiple-comparison significance claims are made.

Conclusion

The directional improvement hypothesis was not supported in this seed: main accuracy did not exceed reward-only ES. Multi-seed studies and archive ablations require a separately frozen amendment after this result is published.

Appendix: reproducibility and failures

Run uv sync --locked, uv run pytest, uv run novelty-es prepare, uv run novelty-es download, and uv run novelty-es run. Repeating run resumes the same clock. uv run novelty-es report regenerates tables, figures, this paper and the static website. Atomic checkpoints contain weights, all RNG streams, archive, cursor, config and deadline. Recorded technical failures: 1. Model-execution elapsed seconds: 20151.339. Final auditing, report preparation and publication follow model execution within the original allowance. Exact exceptions and incomplete conditions are retained in the generated Markdown report and JSON artifacts.

Execution provenance: NVIDIA GeForce RTX 5070 Laptop GPU; Python 3.12.13. Initial execution revision: ddcd928c1392fa2eb5beaff5b7b6f192dfa04817. Training implementation revision: 9f8d30ef0103e3532ebefeafe7f55b8f8d163a80. Frozen generations per arm: 19; inference batch: 16. The timing rule uses the slowest calibration candidate, a complete update and checkpoint, and amortized validation cost. The budget reserves a fraction of each training phase and freezes the common count before training.

Embedding revision: 1110a243fdf4706b3f48f1d95db1a4f5529b4d41. ExTra upstream revision: ea53e68036356f8f276e261e886fbbc102633be3. Complete dependency versions, dataset hashes, absolute deadline, per-phase timings, token-work accounting, raw outputs and completion flags are machine-readable in the published

experiment artifacts. Full-weight noise is regenerated by a hash of perturbation seed and tensor name using an explicit PyTorch generator. Bit-level replay is scoped to the recorded hardware and software environment.

The observed storage-failure recovery regenerated a batch of 16 sampled responses. Decoded text, token IDs, identities and verifier fields matched the preserved failed-write snapshot exactly; only elapsed batch time was excluded. The source hash and machine-readable check are published with the artifacts. This is evidence for this evaluation replay, not arbitrary training restarts or environments.

References

1. ExTra

2. Evolution Strategies at Scale

3. Beyond the Best Guess

4. Novelty-seeking ES

5. EVOL-RL

6. DARLING

7. AC/DC

8. Evaluating Large Language Models Trained on Code

Research protocol and source

AI statement

Codex wrote this codebase and paper, and human cleanup and rewriting are in progress. A human provided the exact specification that was executed.